

TOE (Master): Unitary Core + Gate + ABS Readout + Tail
Control + Moral Semigroup + Legacy

0. Non-claim (scope boundary)

This document is a *closed formal system*: all theorems are proven from explicit axioms below. It does *not* claim to solve open millennium problems (RH, P vs NP , Navier–Stokes) in physical reality. Instead, it provides an operator/readout framework that can *host* those problems as stress tests.

I. The Theory of Everything (single spine)

I.1 Axioms (the only inputs)

Axiom 1 (State space). *There exists a complex Hilbert space $\mathcal{H}$ and a norm $\|\cdot\|$ induced by an inner product $\langle \cdot, \cdot \rangle$.*

Axiom 2 (Unitary core evolution). *There exists a strongly continuous family of unitary operators $U(t) : \mathcal{H} \rightarrow \mathcal{H}$ such that $U(t)^*U(t) = I$ and $\psi(t) = U(t)\psi_0$ for an initial state $\psi_0 \in \mathcal{H}$.*

Axiom 3 (Gate (admissibility projection)). *There exists a (possibly nonlinear) admissibility map $G : \mathcal{H} \rightarrow \mathcal{H} \cup \{\perp\}$ satisfying:*

$$G(\psi) = \psi \text{ if } \psi \in \mathcal{H}_{\text{adm}}, \quad G(\psi) = \perp \text{ if } \psi \notin \mathcal{H}_{\text{adm}},$$

for some admissible subset $\mathcal{H}_{\text{adm}} \subseteq \mathcal{H}$.

Axiom 4 (ABS-first readout). *There exists a bounded linear functional $M \in \mathcal{H}^*$ (equivalently a probe $\phi \in \mathcal{H}$) such that*

$$M(\psi) = \langle \phi, \psi \rangle.$$

The observable is the magnitude (ABS-safe):

$$A(\psi) := |M(\psi)| = |\langle \phi, \psi \rangle| \in \mathbb{R}_{\geq 0},$$

with the convention $A(\perp) = 0$.

Axiom 5 (Moral semigroup (irreversibility by domain cut)). *Define the induced evolution*

$$T_t(\psi) := G(U(t)\psi).$$

The family $\{T_t\}_{t \geq 0}$ is a semigroup on $\mathcal{H} \cup \{\perp\}$:

$$T_{t+s} = T_t \circ T_s, \quad T_0 = I,$$

and $\perp$ is absorbing: $T_t(\perp) = \perp$.

Axiom 6 (Legacy functional). *There exists a measurable integrity density $Q : \mathcal{H} \cup \{\perp\} \rightarrow \mathbb{R}_{\geq 0}$ and a legacy functional*

$$\mathcal{L}(t) := \int_0^t Q(T_\tau(\psi_0)) d\tau,$$

with the absorbing rule $Q(\perp) = 0$.

I.2 Derived structure (definitions)

Definition 1 (Pipeline state and measured time series). Given ψ_0 , define

$$\psi(t) := U(t)\psi_0, \quad \psi_G(t) := T_t(\psi_0) = G(\psi(t)), \quad A(t) := A(\psi_G(t)).$$

Definition 2 (Zeno-enforced gate (optional strengthening)). If G is an orthogonal projection Π_{adm} onto a closed subspace $\mathcal{H}_{\text{adm}}$, define

$$U_Z(t) := \lim_{n \rightarrow \infty} (\Pi_{\text{adm}} U(t/n) \Pi_{\text{adm}})^n$$

whenever the strong limit exists.

I.3 Theorems (unconditional inside this system)

Lemma 1 (Norm invariance inside the unitary core). *For all t , $\|\psi(t)\| = \|\psi_0\|$.*

Proof. $\|\psi(t)\| = \|U(t)\psi_0\| = \|\psi_0\|$ since $U(t)$ is unitary. $\square$

Lemma 2 (ABS-phase invariance (branch safety)). *For any global phase $\alpha \in \mathbb{R}$ and any $\psi \in \mathcal{H}$,*

$$A(e^{i\alpha}\psi) = A(\psi).$$

Proof. $A(e^{i\alpha}\psi) = |\langle \phi, e^{i\alpha}\psi \rangle| = |e^{i\alpha} \langle \phi, \psi \rangle| = |\langle \phi, \psi \rangle| = A(\psi)$. $\square$

Proposition 1 (Semigroup property is enforced by the gate). *For $t, s \geq 0$, $T_{t+s} = T_t \circ T_s$ and $\perp$ is absorbing.*

Proof. This is exactly Axiom 5. The point is: irreversibility is not injected as “energy”, but as an admissibility cut. Once mapped to $\perp$, composition preserves $\perp$. $\square$

Theorem 1 (Legacy monotonicity). *For $t_2 \geq t_1 \geq 0$, $\mathcal{L}(t_2) \geq \mathcal{L}(t_1)$. Moreover, if the trajectory hits $\perp$ at some time t_* , then for all $t \geq t_*$, $\mathcal{L}(t) = \mathcal{L}(t_*)$.*

Proof. By definition,

$$\mathcal{L}(t_2) - \mathcal{L}(t_1) = \int_{t_1}^{t_2} Q(T_\tau(\psi_0)) d\tau \geq 0$$

since $Q \geq 0$. If $T_\tau(\psi_0) = \perp$ for all $\tau \geq t_*$, then $Q(\perp) = 0$ so the integral adds zero past t_* . $\square$

II. Explicit tail control (mechanical stability layer)

II.1 Axioms for a bank kernel

Axiom 7 (Bank kernel decomposition). *There exist atoms $K_j : \mathbb{R} \rightarrow \mathbb{C}$ and weights $w_j^2 \geq 0$ such that*

$$G_{\text{bank}}(\Delta) = \sum_{j=0}^J w_j^2 K_j(\Delta).$$

Axiom 8 (Polynomial decay majorant). *Fix an integer $N \geq 4$. For each j there exists $C_{j,N} > 0$ such that for all $\Delta \in \mathbb{R}$,*

$$|K_j(\Delta)| \leq C_{j,N}(1 + |\Delta|)^{-N}.$$

Axiom 9 (Leakage functional bound). *There exists a functional $\theta(t; J) \geq 0$ and $G_{\text{bank}}(0) > 0$ such that*

$$\theta(t; J) \leq \frac{(\log t) T(J)}{G_{\text{bank}}(0)},$$

where

$$T(J) := \sum_{m \geq 1} (m+1) \sup_{|\Delta| \in [m, m+1]} |G_{\text{bank}}(\Delta)|.$$

II.2 The explicit bound (fully proven)

Lemma 3 (Majorant for G_{bank}). *Let $S_J := \sum_{j=0}^J w_j^2 C_{j,N}$. Then for all $\Delta \in \mathbb{R}$,*

$$|G_{\text{bank}}(\Delta)| \leq S_J(1 + |\Delta|)^{-N}.$$

Proof. Triangle inequality and Axioms II.1–II.2:

$$|G_{\text{bank}}(\Delta)| \leq \sum_{j=0}^J w_j^2 |K_j(\Delta)| \leq \sum_{j=0}^J w_j^2 C_{j,N} (1 + |\Delta|)^{-N} = S_J(1 + |\Delta|)^{-N}.$$

□

Theorem 2 (Explicit tail constant and checkable stability inequality). *With S_J as above,*

$$T(J) \leq (\zeta(N-1) - 1) S_J.$$

Consequently,

$$\theta(t; J) \leq \frac{(\log t)(\zeta(N-1) - 1)}{G_{\text{bank}}(0)} S_J.$$

In particular, the sufficient condition $\theta(t; J) < 1$ is guaranteed if

$$S_J < \frac{G_{\text{bank}}(0)}{(\log t)(\zeta(N-1) - 1)}.$$

Proof. By the majorant lemma, for $|\Delta| \in [m, m+1]$ we have $1 + |\Delta| \geq m+1$, hence

$$\sup_{|\Delta| \in [m, m+1]} |G_{\text{bank}}(\Delta)| \leq S_J(m+1)^{-N}.$$

Therefore

$$T(J) \leq \sum_{m \geq 1} (m+1) S_J(m+1)^{-N} = S_J \sum_{m \geq 1} (m+1)^{-(N-1)}.$$

Reindex $k = m+1$ to get $\sum_{k \geq 2} k^{-(N-1)} = \zeta(N-1) - 1$. Combine with the leakage axiom. □

III. Chaos & harmony co-exist (where each one lives)

Definition 3 (Three placements of irregularity). Irregular/chaotic-looking outputs are permitted without breaking the unitary core by: (i) nonlinear readout/coarse-graining $A(\psi) = |M(\psi)|$, (ii) time-dependent self-adjoint generator (unitary per realization), (iii) open-system Lindblad evolution on density matrices.

Remark 1. In this TOE spine, “harmony” is the internal conservation envelope (unitarity + tail control), while “chaos” is a property of the readout / drive / openness channel. They are orthogonal concerns.

IV. Stress tests (RH / SAT / Navier–Stokes) *as embeddings*, not solved claims

IV.1 RH as an interference-node diagnostic

We can define ABS-safe observables built from prime-log frequencies (no phase unwrapping at nodes). This supports numerics and stability, but it is *not* a proof of RH.

IV.2 SAT as witness interface

SAT is a canonical instance where verification (readout) can be easy while discovery (search in latent space) can be hard. This explains the asymmetry structurally, not as a class separation proof.

IV.3 Navier–Stokes as a Gate/regularity problem

Regularity is precisely the question whether the admissible envelope stays invariant for all time under nonlinear flow. Again: this TOE spine is a framing tool, not a completed millennium proof.

V. Implementation contract (what code is allowed to do)

Definition 4 (Implementation contract). Any code is only allowed to implement: (1) unitary (or explicitly non-unitary) evolution in the model space, (2) the gate G (admissibility; absorbing state), (3) ABS-first readout A , (4) explicit tail checks ($\theta(t; J) < 1$ condition), (5) legacy tracking $\mathcal{L}(t)$. No code is allowed to claim energy creation; outputs are readout scalars / witnesses / diagnostics.

VI. Minimal pseudo-code skeleton (for later)

```
psi = psi0
for step in time:
    psi = U(step) @ psi
    psi = Gate(psi)           # may map to BOT
    A    = abs(<phi, psi>)     # ABS readout (A=0 if BOT)
    L    += Q(psi) * dt        # legacy monotone, Q(BOT)=0
    check_theta()             # tail/leakage inequality
```